Hosting Capacity and the Data-Center Connection Game

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October 19, 2026

An unprecedented surge in demand for electricity

- **Scale.** ERCOT is tracking more than 438 GW of connection requests from large loads as of mid-2026, nearly 90% of them data centers, against an all-time peak demand of 85.5 GW set on August 2023.[1]
- **Single sites are now GW-sized.**
  - OpenAI's Stargate: nearly 7 GW planned across its announced sites.[2]
  - **ANTHROP\C**: $50B of new data centers in Texas and New York;[3] up to 5 GW of compute with Amazon;[4] single training sites of 2 GW (2027) and 5 GW (2028) expected.[5]

[1] [1] ERCOT, *Trending Topics: ERCOT's New Batch Connection Process for Large Electricity Users*, 2026.

[2] [2] OpenAI, *OpenAI, Oracle, and SoftBank Expand Stargate with Five New AI Data Center Sites*, 2025.

[3] [3] Anthropic, *Anthropic Invests $50 Billion in American AI Infrastructure*, 2025.

[4] [4] Anthropic, *Anthropic and Amazon Expand Collaboration for up to 5 Gigawatts of New Compute*, 2026.

[5] [5] Anthropic, *Build AI in America: Anthropic Energy Report*, 2025.

Who decides, and the two questions for today

- In ERCOT, the transmission service provider (e.g., Oncor) and ERCOT study each large-load request.
- Since 2026, ERCOT studies eligible large loads *together* and allocates transmission capacity "based on what the grid can reliably support".[6]

1. How much load can a bus take? — Part I: hosting capacity
2. When requests compete, who gets served? — Part II: the connection game

[6] [1] ERCOT, *Trending Topics: ERCOT's New Batch Connection Process for Large Electricity Users*, 2026.

Part I: Hosting capacity

What is hosting capacity?

Hosting capacity of a location

The largest amount of new *load* that can be connected there without violating the network's operating limits, with the existing network and controls.

- The idea started with *distributed generation*, e.g. rooftop solar photovoltaic (PV) panels on distribution feeders: how much PV can a feeder absorb?[7]
- The same question applies to large new consumption,[8] and to transmission-level loads such as data centers.[9]
- Computed by stochastic simulation, streamlined rules, time series, or optimization (today).[10]

[7] [6] Bollen et al., *Integration of Distributed Generation in the Power System*, 2011, [7] Rylander et al., EPRI, 2015.

[8] [8] Bollen et al., *Energies*, 2017.

[9] [9] Cicilio et al., *IEEE Transactions on Sustainable Energy*, 2021, [10] Wan et al., arXiv:2610.18405, 2026.

[10] [11] Mulenga et al., *International Journal of Electrical Power & Energy Systems*, 2020, [12] Ismael et al., *Renewable Energy*, 2019.

Single-bus hosting capacity as a linear program

Extra load L at bus k in hour t; existing generators redispatch:[11]

$$\mathrm{HC}_{k,t} = \max_{L,\,\mathbf{p}^{\mathrm{g}},\,\mathbf{p},\,\mathbf{f}} \; L$$

$$\text{s.t.} \quad \underbrace{\mathbf{p} = C\,\mathbf{p}^{\mathrm{g}} - \mathbf{d}_t - L\,\mathbf{e}_k}_{\text{net injections (new load at } k)}, \qquad \underbrace{\mathbf{1}^{\top}\mathbf{p} = 0}_{\text{system balance}}, \qquad \underbrace{\mathbf{f} = H\,\mathbf{p}}_{\text{DC power flow}}$$

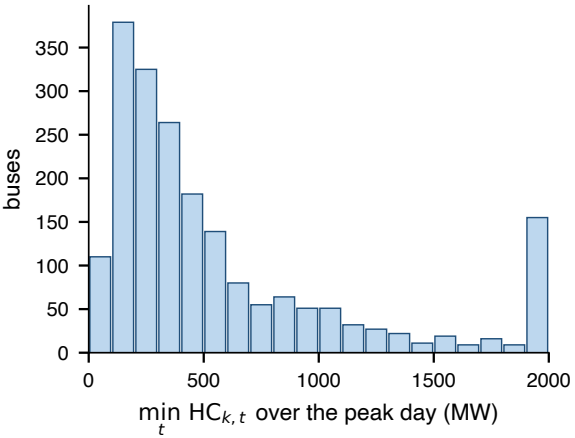
$$\underbrace{|\mathbf{f}| \le \mathbf{F}}_{\text{line thermal limits}}, \qquad \underbrace{\mathbf{p}^{\min} \le \mathbf{p}^{\mathrm{g}} \le \mathbf{p}^{\max}}_{\text{generator limits}},$$

$$L \ge 0$$

Decision variables: L new load, $\mathbf{p}^{\mathrm{g}}$ generator outputs, $\mathbf{p}$ net bus injections, $\mathbf{f}$ line flows. Data: C maps generators to buses, $\mathbf{d}_t$ bus loads, H the PTDF (shift-factor) matrix, $\mathbf{F}$ line limits.

[11] [13] Alturki et al., *Applied Energy*, 2018, [10] Wan et al., arXiv:2610.18405, 2026.

Hosting capacity varies a lot across buses

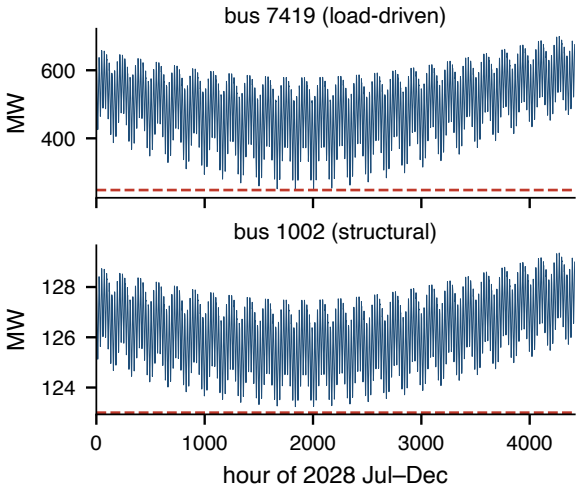


ACTIVSg2000 synthetic Texas grid,[a] 2028 Jul–Dec, the 24 hours of the peak-load day, all 2,000 buses:

- median 372 MW
- 10% of buses below 130 MW
- 90% below 1,536 MW; 146 buses above 2,000 MW (not shown)
- smallest: 10 MW

[a] [14] Birchfield et al., *IEEE Transactions on Power Systems*, 2017.

Over time: the stressed hours bind



A constant load must be served in *every* hour:

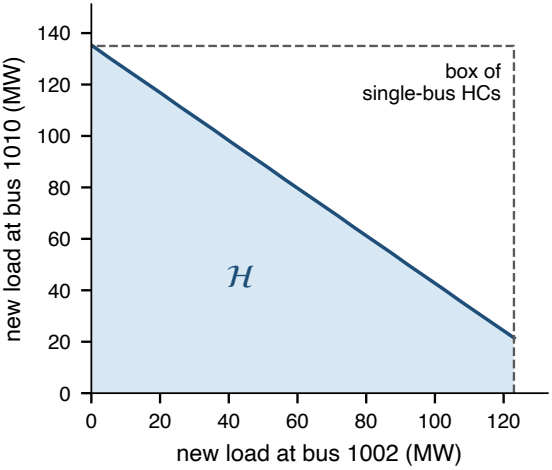
$$\mathrm{HC}_k = \min_t \mathrm{HC}_{k,t} \quad \text{(dashed)}$$

- Bus 7419: 248 MW to 699 MW, median 503 MW: tight at peak hours.
- Bus 1002: 123 MW to 129 MW: the network limits it, not the load cycle.

"Dynamic hosting capacity" treats this time variation explicitly.[a]

[a] [15] Oliveira et al., *Energies*, 2019.

Many buses at once: the hosting-capacity region



$$\mathcal{H} = \left\{ \mathbf{x} \geq 0 : \begin{array}{c} \forall t \ \exists \mathbf{p}_t^{\mathrm{g}}, \mathbf{p}_t : \\ \mathbf{p}_t = C\,\mathbf{p}_t^{\mathrm{g}} - \mathbf{d}_t - \mathbf{x}, \quad \mathbf{1}^\top \mathbf{p}_t = 0, \\ |H\mathbf{p}_t| \leq \mathbf{F}, \quad \mathbf{p}^{\min} \leq \mathbf{p}_t^{\mathrm{g}} \leq \mathbf{p}^{\max} \end{array} \right\}$$

The LP's constraints with new loads $\mathbf{x}$ (one per bus) in place of $L\,\mathbf{e}_k$.

- A convex polytope: the projection of the DCOPF feasible set.[a]
- Single-bus HCs are its axis intercepts; they do not add up.[b]
- Peak hour: bus 1002 hosts 123 MW alone, only 81 MW with 60 MW at neighbor 1010.

[a] [16] Gong et al., *Electric Power Systems Research*, 2024.

[b] [10] Wan et al., arXiv:2610.18405, 2026.

With a local generator

Site at bus k: load L, private generator $0 \le g_t \le G$. The grid only sees the *net* withdrawal $L - g_t$:

$$\text{feasible} \iff \exists\, g_t \in [0, G] : \underbrace{L - g_t}_{\text{net withdrawal}} \le \mathrm{HC}_{k,t} \ \ \forall t \implies \underline{\mathrm{HC}_k^G = \min_t \mathrm{HC}_{k,t} + G.}$$

- Exact when the generator is always available; with outages, ramping, or fuel limits, $\mathrm{HC}_k^G \le \mathrm{HC}_k + G$.
- The generator runs only when the grid is short or when its cost is below the local price.

With storage: the hours become coupled

Battery at bus k: power P, energy B, signed output u_t (positive = discharge), time step Δt; $\mathrm{HC}_{k,t}$ comes from the single-bus LP:

$$\mathrm{HC}_k^S = \max_{L,\,u,\,e} \; L \quad \text{s.t.} \qquad \underbrace{L - u_t \le \mathrm{HC}_{k,t}}_{\text{grid limit on net withdrawal}}, \qquad \underbrace{-P \le u_t \le P}_{\text{power}},$$

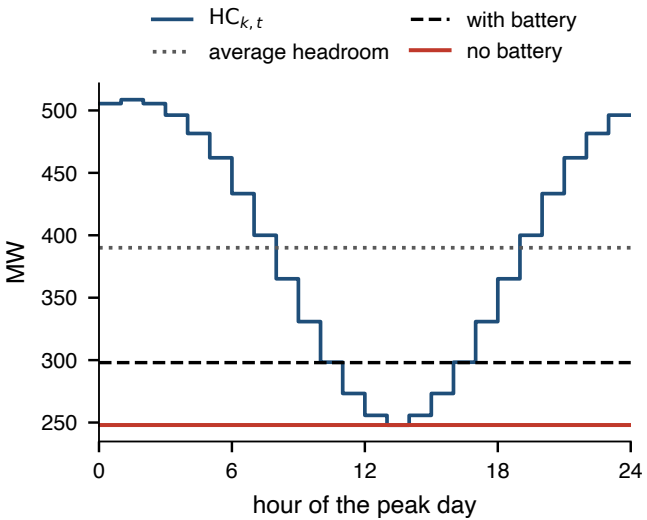
$$\underbrace{e_t = e_{t-1} - \Delta t\, u_t, \quad 0 \le e_t \le B}_{\text{energy}}, \qquad \underbrace{e_0 = e_T}_{\text{daily cycle}}.$$

Decision variables: L new load, u_t battery output, e_t stored energy.

- A multi-period LP: discharge in tight hours, recharge in slack ones.[12]
- Summing over the cycle ($\sum_t u_t = 0$): $\mathrm{HC}_k^S \le \frac{1}{T}\sum_t \mathrm{HC}_{k,t}$, so storage lifts the limit at most from the minimum to the average headroom.
- Power ($L \le \mathrm{HC}_{k,t} + P$) and energy ($B$) keep it below that.

[12] [17] Macedo et al., *IEEE PES Innovative Smart Grid Technologies Conference (ISGT)*, 2017.

Storage over one day

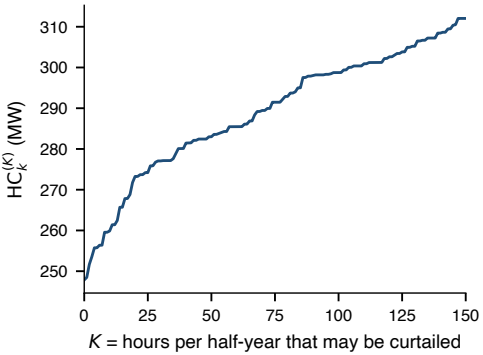


Bus 7419, peak day, hourly steps, daily cycle ($e_0 = e_{24} = B/2$), $B = 4P$:

- no battery: 248 MW
- average headroom (dotted): 390 MW, the upper limit
- $P = 50\,\mathrm{MW}$, $B = 200\,\mathrm{MWh}$: **298** MW

B limits how long a shortfall the battery can cover. Half-year: 248 $\rightarrow$ 298 MW.

Flexibility: curtail in a few hours



Flexible connection:[a]

$$\max_{L,\,\delta} \; L$$

$$\text{s.t. } L - \delta_t \le \mathrm{HC}_{k,t}, \quad \delta \ge 0, \quad \|\delta\|_0 \le K$$

- δ_t is the load curtailed in hour t; nonconvex ($\|\delta\|_0$ counts curtailed hours).
- $\mathrm{HC}_k^{(K)} = \mathrm{HC}_{k,[K+1]}$, the $(K+1)$-th *smallest* hourly value.
- Bus 7419: 248 MW at $K = 0$, 274 MW at $K = 24$ (+11%), 299 MW at $K = 100$ (+21%). Diminishing returns in K.[b]

[a] [18] Gu et al., *Hawaii International Conference on System Sciences (HICSS)*, 2026.

[b] [19] Etherden et al., *IEEE PowerTech*, 2011, [20] Sarker et al., arXiv:2604.20185, 2026.

How much should we trust a hosting-capacity number?

- **Security.** Dropping N-1 contingencies overstates data-center hosting capacity by about 20% on the IEEE RTS-24.[13]
- **Dynamics.** The data center's own ride-through limits (UPS, converters) can be the binding constraint.[14]
- **Model.** DC power flow has no voltage or reactive-power limits and no losses.
- **Simultaneity.** Single-bus numbers ignore everyone else who is connecting at the same time.

The last point leads to the second question: who gets served?

[13] [10] Wan et al., arXiv:2610.18405, 2026.

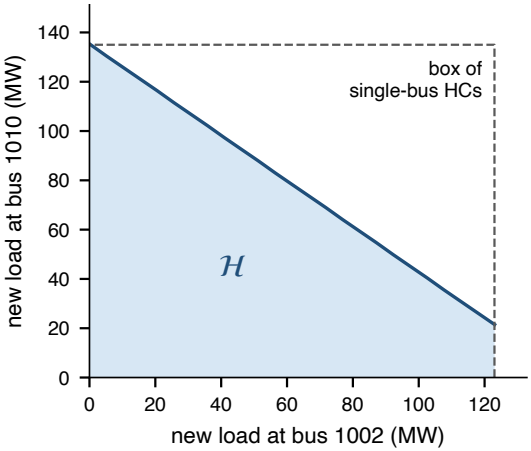
[14] [21] Ren et al., arXiv:2609.03030, 2026.

Part II: The connection game

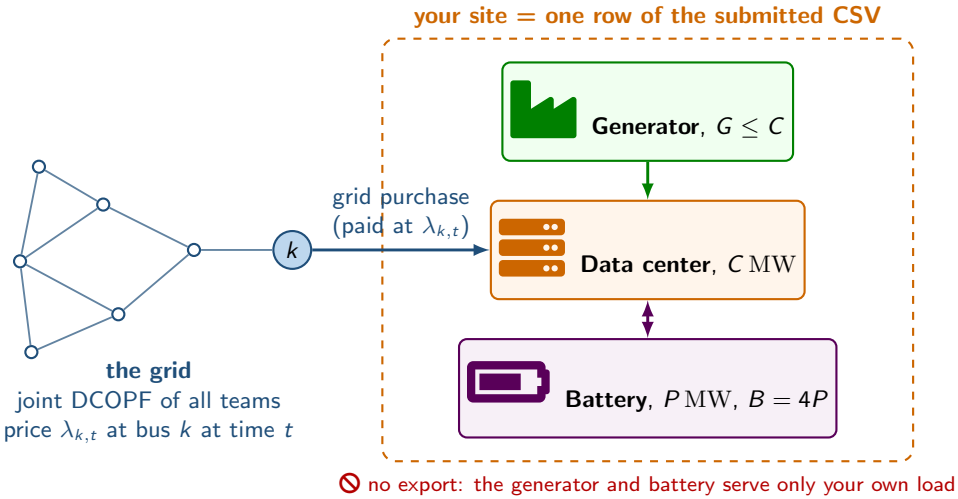
From one bus to many applicants

- Part I: one bus $\Rightarrow$ one number; many buses $\Rightarrow$ a region.
- Two requests can each fit alone and still not fit together.
- Someone has to decide who is served, and by how much.

For the next three weeks, you are the applicants.



Your role in one picture

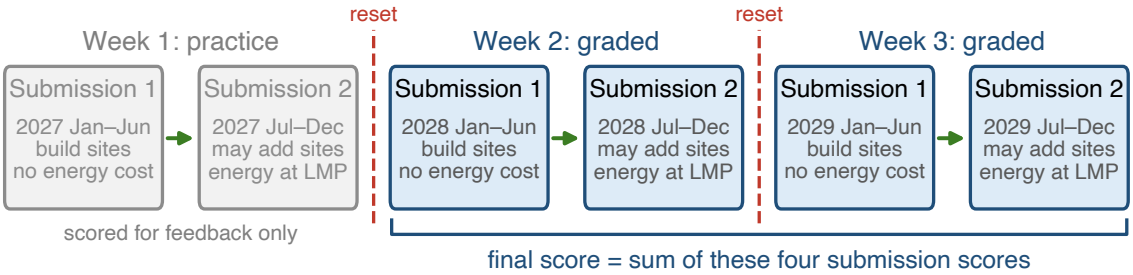


You choose the numbers C, G, P for each site, and the grid operator runs them.

What earns money, and when

One 50 MW site built in Submission 1

- earns $6,500 per MWh served and costs $25M per MW (paid once).
- Energy is about $18/MWh.
- It breaks even at about 44% served over a week, so what matters is whether the grid can deliver.



What the grid operator does

Part I's LP, with every team's submitted plans:

$$
\begin{aligned}
&\min \text{ generation cost} + \text{private generator cost} \\
&\text{s.t. } \mathbf{p} = C\mathbf{p}^{\mathrm{g}} - \mathbf{d}_t - (\text{all teams' sites, net of their own generator and battery}), \\
&\qquad \mathbf{1}^{\top}\mathbf{p} = 0, \quad \mathbf{f} = H\mathbf{p}, \quad |\mathbf{f}| \le \mathbf{F}, \quad \text{batteries' constraints.}
\end{aligned}
$$

- One DCOPF per six-hour period, coupled through the batteries over each day.
- The price of energy at your bus is the LMP.

When it doesn't fit: how ERCOT connects large loads since 2026

Before: one request at a time

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- Repeated restudies; delays of years.

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- All eligible loads ($\geq 75\,\mathrm{MW}$) are studied together.
- MW are allocated "based on what the grid can reliably support".

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Minimum allocation[a]

- Below $100\,\mathrm{MW}$, or below 90% of a request of $100\,\mathrm{MW}$ or less, the allocation is reduced to zero.

Ways to connect sooner

- Bring your own generation, or let ERCOT curtail you under stress (Part I).

[a] [22] ERCOT, ERCOT Board of Directors, Item 15.1.4.3, 2026.

The game's batch rule: one shared limit

Three teams behind one line that can carry $200\,\mathrm{MW}$ of new load.
Requests: A 40, B 100, C 200 (total 340).

	request	step 1	share
A	40	40	100%
B	100	80	80%
C	200	80	40%

[15] [23] Sprumont, *Econometrica*, 1991.

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3. **Recompute without C.** $40 + 100 = 140 \leq 200$: A and B are served in full.

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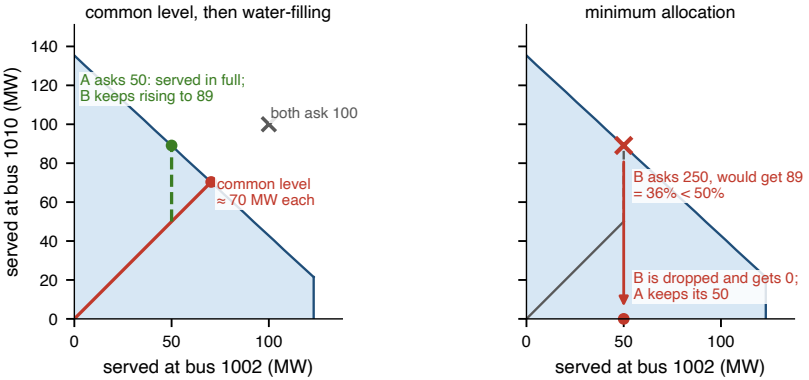
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C still pays capital on 200 MW. Asking for more than the grid can give is the most expensive mistake in the game.

ERCOT does not publish a formula for splitting capacity among studied loads. The game uses equal MW (the uniform rule[15]) so that everyone can predict it.

[15] [23] Sprumont, *Econometrica*, 1991.

On the network: sharing the region



1. Everyone rises to a common MW level. A team that asked for less is served in full.
2. Teams that are not blocked keep rising (water-filling).
3. A team that would get less than half of its request gets nothing.

The game and ERCOT

	ERCOT batch study	This game
Who is studied together	All eligible loads in a batch	All teams' submissions
Already connected	Base load keeps its MW	`ACTIVSg2000`'s existing load
How MW are split	System-wide study	Equal MW level, then water-filling
Too small a share	below $100\,\mathrm{MW}$ (90% for requests $\leq 100\,\mathrm{MW}$) $\to 0$	below 50% of the request $\to 0$
Commitment	Financial security, site control	Capital charged on the full request
Own generation	Offsets transmission need	Private generator G
Flexible load	May be curtailed under stress	Not in the game (Part I)

A two-team example

- Team A at bus 1002.
- Team B at neighbor 1010.
- Each picks 50 or 100 MW, built in Submission 1.

Payoffs per week ($M):

A \ B	B small (50)	B large (100)
A small (50)	1,601 / 1,601	1,601 / 2,580
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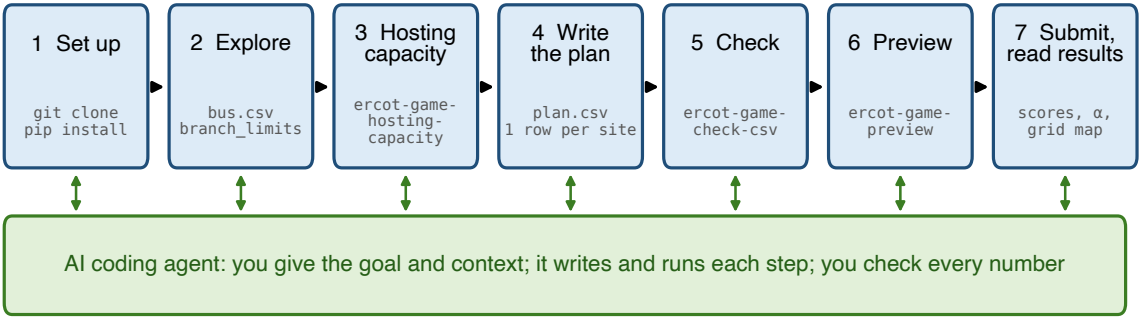
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- Your score depends on where everyone builds, and you see the results between submissions. What would you do?

From the slides to your laptop



Everything is in one GitHub repository: [Link].

An AI coding agent on the IEEE 14-bus demo (1/2)

1. Read the rules

Read the README and Part I of the tutorial, then explain in under 150 words what my team submits, how it is scored, and which commands I will use. Do not change any files.

Agent reply (shortened)

One CSV per team per submission [...] six files, and no report. [...] If they don't fit, everyone gets the same MW level, and a team that would get less than half of its request gets nothing.

Check it yourself

Accurate, but about 190 words: agents do not always respect limits.

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2. Look at the data

Summarize the demo package: buses, lines, generators, total load, periods, and the five lines with the smallest thermal limits. Show the commands you ran.

Agent reply (shortened)

14 buses, 20 lines, 5 generators (772 MW); load 243–288 MW; 168 hourly periods, peak in hour 157. Smallest limits: 12-13 (20), 10-11 (25), 13-14 (25). README tip: buses 13 and 14 share a bottleneck.

Check it yourself

Count the rows in `bus.csv` and `branch_limits.csv`.

3. Hosting capacity

Compute the hosting capacity of every bus over all 168 hours, sort it, explain it in one sentence, and shortlist two buses.

Agent reply (shortened)

2,352 LPs, no failures. Bus 1: 485 MW, bus 2: 441 MW, . . . , bus 14: 39 MW. Shortlist 1 and 2, but "you can't build 485 MW at one and 441 MW at the other", and every team sees this table.

Check it yourself

Re-run one bus yourself. AI agents usually make mistakes that are not easy to see.

An AI coding agent on the IEEE 14-bus demo (2/2)

4. Write and check a plan

One site at each shortlisted bus at about half its hosting capacity; check it. Then break it on purpose (generator larger than connection).

Agent reply (shortened)

240 MW at bus 1, 220 MW at bus 2: `VALID`. The broken copy: `INVALID - row 3: generator capacity cannot exceed connection capacity.` "Valid only means the file is well-formed."

Check it yourself

Open the CSV; breaking a file shows what the checker catches.

5. Preview

Preview it as Submission 2. Report the share served, the score, the dominant term, and whether it beats break-even.

Agent reply (shortened)

100% served. IT value \$502.3M, site cost \$437.5M, energy \$2.3M; score +\$62.5M. Break-even at about 88% served: "a margin of about 12 percentage points."

Check it yourself

Redo $460 \times 168 \times 6{,}500 = \$502.3\text{M}$ by hand.

6. A rival copies you

Simulate a rival with the same plan and run the joint evaluation. What service fraction does each team get? List every assumption I should check.

Agent reply (shortened)

Both get $\alpha = 0.527$: 242 of 460 MW. Score −\$174M instead of +\$62.5M, just above the 50% cutoff. Among 11 assumptions: "I didn't consider what other teams would do."

Check it yourself

Open `evaluation.json`; always ask for the assumptions.

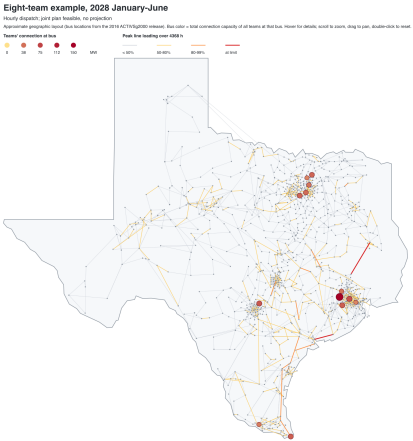
After each round, and your first plan

What you get back

- your score breakdown (in private);
- every team's service fraction $\alpha\%$;
- a grid map with the total MW at each bus.

A first plan for Week 1

1. Shortlist buses by hosting capacity.
2. Ask well below it if w/o a generator or battery.
3. Check, preview, change one thing, compare.



Eight-team example: bus color = total MW of all teams, line color = peak loading; approximate locations.

Takeaways

- Hosting capacity is an LP: the DCOPF with a "maximize the new load" objective.
- Time: a constant load is limited by the worst hour. Generation adds G and storage moves the limit toward the average. Flexibility toward the $(K+1)$-th worst.
- Space: single-bus numbers are intercepts of a region and do not add up.
- With many applicants, who gets served is a design choice, and it turns planning into a game.

Questions for you. How should a grid operator allocate scarce headroom among data centers? What would you change in the game's rule?

References I

[1] ERCOT. *Trending Topics: ERCOT's New Batch Connection Process for Large Electricity Users*. June 2026.

[2] OpenAI. *OpenAI, Oracle, and SoftBank Expand Stargate with Five New AI Data Center Sites*. Sept. 2025.

[3] Anthropic. *Anthropic Invests $50 Billion in American AI Infrastructure*. Nov. 2025.

[4] Anthropic. *Anthropic and Amazon Expand Collaboration for up to 5 Gigawatts of New Compute*. Apr. 2026.

[5] Anthropic. *Build AI in America: Anthropic Energy Report*. July 2025.

[6] M. H. J. Bollen and F. Hassan. *Integration of Distributed Generation in the Power System*. Wiley-IEEE Press, 2011.

[7] M. Rylander, L. Rogers, and J. Smith. *Distribution Feeder Hosting Capacity: What Matters When Planning for DER?* Tech. rep. 3002004777. EPRI, 2015.

[8] M. Bollen and S. Rönnberg. "Hosting Capacity of the Power Grid for Renewable Electricity Production and New Large Consumption Equipment". In: *Energies* 10.9 (2017), p. 1325.

[9] P. Cicilio et al. "Transmission Hosting Capacity of Distributed Energy Resources". In: *IEEE Transactions on Sustainable Energy* 12.2 (2021), pp. 794–801.

[10] H. Wan et al. *Impact of Security Modeling Fidelity on Hyperscale Data Center Hosting Capacity Assessment*. arXiv:2610.18405. 2026.

[11] E. Mulenga, M. H. J. Bollen, and N. Etherden. "A Review of Hosting Capacity Quantification Methods for Photovoltaics in Low-Voltage Distribution Grids". In: *International Journal of Electrical Power & Energy Systems* 115 (2020), p. 105445.

[12] S. M. Ismael et al. "State-of-the-Art of Hosting Capacity in Modern Power Systems with Distributed Generation". In: *Renewable Energy* 130 (2019), pp. 1002–1020.

[13] M. Alturki et al. "Optimization-Based Distribution Grid Hosting Capacity Calculations". In: *Applied Energy* 219 (2018), pp. 350–360.

[14] A. B. Birchfield et al. "Grid Structural Characteristics as Validation Criteria for Synthetic Networks". In: *IEEE Transactions on Power Systems* 32.4 (2017), pp. 3258–3265.

References II

[15] T. E. C. de Oliveira et al. "The Concept of Dynamic Hosting Capacity for Distributed Energy Resources: Analytics and Practical Considerations". In: *Energies* 12.13 (2019), p. 2576.

[16] S. Gong and J. F. G. Cobben. "Adaptive Facet Selection in Multidimensional Hosting Capacity Region Assessment". In: *Electric Power Systems Research* 234 (2024), p. 110654.

[17] L. H. Macedo et al. "Increasing the Hosting Capacity for Renewable Energy in Distribution Networks". In: *IEEE PES Innovative Smart Grid Technologies Conference (ISGT)*. 2017.

[18] N. Gu, G. Chen, and J. Qin. "The Role of Flexible Connection in Accelerating Load Interconnection in Distribution Networks". In: *Hawaii International Conference on System Sciences (HICSS)*. arXiv:2510.11476. 2026.

[19] N. Etherden and M. H. J. Bollen. "Increasing the Hosting Capacity of Distribution Networks by Curtailment of Renewable Energy Resources". In: *IEEE PowerTech*. Trondheim, 2011.

[20] G. C. Sarker and N. Dahlin. *Risk-Aware Hosting Capacity Analysis for Flexible Load Interconnection in Distribution Networks*. arXiv:2604.20185. 2026.

[21] P. Ren, W. Sun, and F. Teng. *Hosting Capacity Assessment of Data Centers with Voltage Ride-Through Capability in Power Systems*. arXiv:2609.03030. 2026.

[22] ERCOT. *ERCOT Comments on NPRR1325 and PGRR145*. ERCOT Board of Directors, Item 15.1.4.3. May 2026.

[23] Y. Sprumont. "The Division Problem with Single-Peaked Preferences: A Characterization of the Uniform Allocation Rule". In: *Econometrica* 59.2 (1991), pp. 509–519.

Backup: the evaluator in full

For each day d and period $t = 1, \dots, 4$ ($\Delta t = 6\,\text{h}$): a DCOPF with all teams' sites, coupled through each battery's daily cycle.

$$\min \ \sum_{d,t} \Delta t \big(\mathbf{c}^\top \mathbf{p}^{\text{g}}_{dt} + \sum_s \sum_m c^m q^m_{s,dt} \big)$$

s.t. DCOPF balance and line limits with loads $\alpha_i C_s$ and private q_s, u_s

$$e_{s,d,t} = e_{s,d,t-1} - \Delta t \, u_{s,dt}, \quad 0 \le e \le B_s, \quad |u| \le P_s, \quad e_{s,d,0} = e_{s,d,4} = B_s/2$$

$$\sum_{s \in i,\, k(s)=n} u_{s,dt} \le \sum_{s \in i,\, k(s)=n} \alpha_i C_s \quad \text{(a team's assets serve only its own load)}$$

- An exact LP that separates by day (Benders or one LP per day).
- LMP = dual of nodal balance; it prices energy in Submission 2.